

CE 310 — Probability and Statistics for Civil and Architectural Engineering

Policy & Syllabus — Fall 2026

Course Information

Course number	CE 310
Credits	3 units
Meeting pattern	Three 50-minute sessions per week
Session 1 (Lecture)	Theory, case studies, interactive problems
Session 2 (Tutorial)	Guided tool walkthrough — Excel (Weeks 1–8), Python/Colab (Weeks 9–15)
Session 3 (Lab)	Hands-on practice with answer sheet or notebook
Prerequisite	MATH 129 (Calculus II)
Enrollment	Civil Engineering and Architectural Engineering students

Instructor

Instructor	Dr. Wooyoung Jung
Office	Civil Building, Room 220D
Phone	+1 (520) 621-7469
Email	wooyoung@arizona.edu
Email response time	Within 48 hours on business days
Office hours	By appointment — email wooyoung@arizona.edu to schedule
Course page	D2L

Course Description

An applied introduction to probability, statistics, and data analysis for civil and architectural engineering. Students learn to characterize data distributions, apply probability models to engineering design decisions, build regression and simulation models, and communicate uncertainty in building performance analysis. The course develops both conceptual understanding — why probabilistic reasoning is essential for safe and efficient design — and practical skills using industry-standard tools (Microsoft Excel and Python).

Weeks 1–8 develop statistical foundations through Excel analysis of real climate and building energy datasets. Weeks 9–15 extend these foundations into predictive modeling, hypothesis testing,

sensitivity analysis, simulation, and uncertainty quantification using Python (Google Colab). The course closes with a capstone probabilistic design report and presentation.

Learning Outcomes

Upon successful completion of CE 310, students will be able to:

1. **Characterize distributions** — compute and interpret descriptive statistics (mean, median, mode, standard deviation, IQR, skewness), histograms, box plots, and design percentiles for engineering datasets.
 2. **Apply probability fundamentals** — use classical probability rules, conditional probability, and Bayes' theorem to evaluate engineering decisions under uncertainty.
 3. **Model with distributions** — select and apply discrete (Poisson) and continuous (Normal) probability distributions to engineering phenomena; compute probabilities, return periods, and design thresholds.
 4. **Compare empirical and parametric models** — evaluate the fit of parametric distributions against observed data and choose the appropriate approach for design applications.
 5. **Build and interpret regression models** — develop simple and multiple linear regression models; assess model significance with t-tests and ANOVA; detect confounding through conditional analysis.
 6. **Quantify uncertainty by simulation** — design Monte Carlo simulations to propagate input uncertainty to output distributions; construct and interpret bootstrap confidence intervals.
 7. **Communicate probabilistic design recommendations** — translate statistical analysis into engineering memos, reports, and presentations addressed to a technical client audience.
 8. **Use professional analysis tools** — apply Excel and Python (Google Colab with pandas, scipy, statsmodels, matplotlib) to analyze real building performance datasets at professional scale.
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Required Materials

Textbook: None required. All lecture notes, datasets, lab instructions, and reference materials are provided through D2L.

Software: - Microsoft Excel — available on all University of Arizona computer lab machines (no personal installation required) - Python via Google Colab — browser-based; no local installation required. Students need a Google account.

Hardware: A laptop or tablet capable of running a web browser is sufficient for all lab sessions.

Reference texts (optional — not required for exams): - Ang & Tang, *Probability Concepts in Engineering Planning and Design*, John Wiley - Haldar & Mahadevan, *Probability, Reliability and Statistical Methods in Engineering Design*, John Wiley - Bruce, Bruce & Gedeck, *Practical Statistics for Data Scientists*, O'Reilly

Grading

Component	Weight	Notes
Pre-Lab Quizzes	10%	10 multiple-choice questions; due on D2L before Session 3 (Lab) each week (Weeks 1–7, 9–14)
Lab Assignments	25%	Due before the following week's Session 1; auto-graded components + manual memo/chart rubric
Midterm Exam	20%	Week 8: Session 1 — Concept Exam (50 min, closed-computer); Session 2 — Lab Practicum (50 min, open Excel, open notes)
Final Exam	25%	Week 15: Session 1 — Written Methods Exam (open notes, closed computer and AI, covers Weeks 9–14 concepts); Session 2 — Python Practicum (Colab only, no internet or AI, new dataset)
Capstone Project + Presentation	20%	Teams of 3 (instructor-assigned, ~15 teams); each team chooses its own real-world civil/architectural engineering dataset; written report + 4-min presentation (Dec 7 and Dec 9, last two class days)

Grade scale:

Grade	Range
A	90–100
B	80–89
C	70–79
D	60–69
E	Below 60

Grade cutoffs may be lowered but will not be raised mid-semester. The course is not curved.

16-Week Schedule

Holiday adjustments — three class days are cancelled this semester: - **Week 3 (Sep 7–11):** No class Mon Sep 7 (Labor Day). Lecture ☐ Wed Sep 9; Tutorial ☐ Fri Sep 11; Lab is independent (due before Mon Sep 14). - **Week 12 (Nov 9–13):** No class Wed Nov 11 (Veterans Day). Lecture ☐ Mon Nov 9; Tutorial ☐ Fri Nov 13; Lab is independent (due before Mon Nov 16). - **Week 14 (Nov 23–27):** No class Fri Nov 27 (Thanksgiving). Lecture ☐ Mon Nov 23; Tutorial ☐ Wed Nov 25; Lab is independent (due before Mon Nov 30).

Part I — Statistical Foundations (Excel, Weeks 1–8)

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
1 Aug 24, 26, 28	The Flaw of Averages	Deterministic vs. probabilistic design. Mean, median, mode, range, standard deviation, variance, quartiles, IQR. Design percentile concept (ASHRAE 1% dry-bulb). Case: <i>Texas Winter Storm Uri — grid sized to average, left tail killed people.</i>	Tucson 2023 hourly weather (8,760 rows). Compute descriptive statistics. Find 99th-percentile design temperature with =LARGE(). Chart: time-series temperature vs. annual mean line.	Lab Assignment 1 + Pre-Lab Quiz 2

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
2 Aug 31, Sep 2, 4	Shape Matters	Histograms: bins, frequency, shape. Skewness: definition, SKEW coefficient, three shapes. Box plots: five-number summary, IQR, outlier fences. Percentile formalization: <code>=PERCENTILE()</code> . <i>Case: Wind pressure scales as V^2 — average wind grossly underestimates peak loads.</i>	Build temperature, wind, and humidity histograms. Compute SKEW() for all three variables. Build box-and-whisker chart. Interpret shape, identify skewed variables.	Lab Assignment 2 + Pre-Lab Quiz 3
3 Sep 9, 11	Probability and Bayes' Theorem	Classical probability: sample space, events, complement, addition rule. Conditional probability: $P(A B)$ via COUNTIFS. Independence. Bayes' theorem: $P(B A) = P(A B) \cdot P(B) / P(A)$. <i>Case: HVAC fault detection — given anomalous energy reading, what is $P(\text{equipment fault})$? Base rates matter.</i>	Tucson office energy dataset (8,760 rows: cooling, heating, lighting). Compute $P(\text{high cooling} \text{summer})$, $P(\text{summer} \text{high cooling})$ via COUNTIFS. Verify Bayes numerically. Engineering decision: justify cooling system sizing from conditional probabilities. (Lab independent — no Sep 7 class. Due before Mon Sep 14.)	Lab Assignment 3 + Pre-Lab Quiz 4

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
4 Sep 14, 16, 18	Rare Events and Return Periods	Poisson distribution: PMF, CDF, parameter λ . Return period: $T = 1/p$. Exceedance probability. <code>=POISSON.DIST()</code> . <i>Case: Monsoon drainage design — ASCE 7 100-year storm; sizing to average annual rainfall leads to chronic flooding.</i>	Tucson daily precipitation 2013–2022 (3,650 rows). Fit Poisson to monthly rain-day counts. Compute $P(X \geq k)$, $P(X = 0)$, CDF. Calculate 25-year and 100-year return periods. Justify drainage design threshold.	Lab Assignment 4 + Pre-Lab Quiz 5
5 Sep 21, 23, 25	The Normal Distribution	Normal distribution: PDF shape, parameters μ and σ . Standardization and Z-scores. <code>=NORM.DIST()</code> , <code>=NORM.INV()</code> . ASHRAE 55 thermal comfort compliance. <i>Case: Building zone temperature control — normal model for indoor temp; quantify hours above comfort band.</i>	DOE Medium Office zone temperature dataset (8,760 rows). Fit Normal model to zone temperature. Compute $P(T > \text{threshold})$, $P(a < T < b)$. Find 90th and 95th percentile design temperatures. Compute ASHRAE 55 occupied-hours exceedance rate.	Lab Assignment 5 + Pre-Lab Quiz 6

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
6 Sep 28, 30, Oct 2	Empirical vs. Parametric	Empirical distribution: direct counting vs. Normal model. When does parametric fit fail? Percentile comparison: =PERCENTILE() vs. =NORM.INV(). Design standard rationale: why ASHRAE uses empirical tables, not Normal fits, for extreme events. Case: <i>HVAC peak sizing — Normal model underestimates 99th-percentile cooling demand when distribution has a long right tail.</i>	Reuse Week 5 dataset. Compute empirical exceedance $P(\text{outdoor} > 85^{\circ}\text{F})$ and compare to Normal model. Build empirical ASHRAE 55 compliance band. Explain where Normal fits well and where it fails.	Lab Assignment 6 + Pre-Lab Quiz 7

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
7 Oct 5, 7, 9	Correlation and Conditional Analysis	Pearson r: formula, interpretation, limitations. Coefficient of determination r^2 . Conditional correlation: does r change by season or occupancy? Simpson's Paradox: aggregate correlation reverses sign within subgroups. <i>Case: Outdoor temp vs. cooling energy — the signal an HVAC energy model must capture.</i>	Compute CORREL() for full year, summer only, business hours only. Compute r^2 and interpret variance explained. Find a case of apparent Simpson's Paradox in the building dataset. Interpret sign reversal.	Lab Assignment 7 + Pre-Lab Quiz 8
8 Oct 12, 14, 16	Midterm	Mon Oct 12: Review session — connections across Weeks 1–7; common mistakes on histograms, conditional probability, Normal CDF.	Wed Oct 14 (50 min): Closed-computer Concept Exam — definitions, interpretation, short calculation. Fri Oct 16 (50 min): Open-Excel Lab Practicum — new dataset, familiar question types.	—

Part II — Predictive Modeling and Uncertainty (Python / Google Colab, Weeks 9–15)

Python environment: Google Colab (browser-based). No local installation. Key libraries: pandas, numpy, scipy, statsmodels, matplotlib. A 15-minute Python Quickstart notebook is provided at the start of Week 9.

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
9 Oct 19, 21, 23	Simple Regression	Linear model $y = \beta_0 + \beta_1 x$. Least squares. Residuals. Slope, intercept, p-value, r^2 . Interpreting the regression line for engineering prediction. Case: <i>Predicting cooling load from outdoor temperature — the baseline model every energy audit uses.</i>	<code>scipy.stats.linregress</code> on zone temperature vs. cooling energy. Scatter plot with fitted line. Interpret slope (kWh/°F), intercept, r-value. Predict cooling at ASHRAE design temperature. Residual plot. Sidebar: confidence interval for the mean prediction vs. prediction interval for a single observation — why equipment sizing requires the PI, not the CI.	Lab Assignment 9 + Pre-Lab Quiz 10
10 Oct 26, 28, 30	Multiple Regression	Adding predictors. Adjusted R^2 . Interaction terms. Multicollinearity concept. Residual diagnostics. Case: <i>Cooling load from outdoor temp + business hours flag — separating weather signal from occupancy signal.</i>	<code>statsmodels.formula.api</code> . Add BusinessHours and Season as predictors. Full regression table: coefficients, p-values, R^2 , adjusted R^2 . Residual plot for model assumptions. Compare adjusted R^2 vs. simple regression.	Lab Assignment 10 + Pre-Lab Quiz 11

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
11 Nov 2, 4, 6	Hypothesis Testing	Null vs. alternative hypothesis. t-statistic, p-value, $\alpha = 0.05$. Two-sample t-test. One-way ANOVA. Statistical vs. practical significance. <i>Case: A/B test of two HVAC control strategies — does the difference in mean energy use reflect true system performance or random variation?</i>	<code>scipy.stats.ttest</code> Compare mean cooling energy between occupied and unoccupied hours. Visualize with overlapping histograms. Interpret p-value. Discuss statistical significance vs. practical engineering relevance.	Lab Assignment 11 + Pre-Lab Quiz 12
12 □ Nov 9, 13	Sensitivity Analysis	One-at-a-time (OAT) sensitivity. Identifying dominant drivers of uncertainty. Tornado charts. Partial derivatives as sensitivity coefficients. <i>Case: Which input parameter drives annual energy uncertainty the most — outdoor temperature, occupancy schedule, or setpoint?</i>	Vary each input parameter $\pm 20\%$ from baseline; compute output change. Rank inputs by impact. Plot tornado chart with <code>matplotlib.barh</code> . Identify which variable most warrants monitoring. (Lab independent — no Nov 11 class. Due before Mon Nov 16.)	Lab Assignment 12 + Pre-Lab Quiz 13

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
13 Nov 16, 18, 20	Monte Carlo Simulation	Random sampling from distributions. Law of large numbers. Input uncertainty □ output distribution. P10/P50/P90. Case: <i>Probabilistic peak cooling demand under uncertain occupancy and weather — sizing equipment to a distribution, not a point estimate.</i>	<code>numpy.random</code> sampling from Normal inputs (10,000 scenarios). Compute output distribution. Report P10/P50/P90 with <code>numpy.percentile</code> . Histogram of simulated results. Compare to deterministic estimate. Capstone kickoff (end of Fri Nov 20 lab): teams of 3 assigned; each team chooses its own dataset (see <code>Wk15.16_Dataset_Selection_Guide.md</code>) — no topic-approval step required.	Lab Assignment 13 + Pre-Lab Quiz 14

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
14 □ Nov 23, 25	Bootstrap Confidence Intervals	Bootstrap resampling: why it works. Confidence interval construction. Interpreting CI width in engineering context. Connection to regression coefficient uncertainty. <i>Case: How confident are we in the slope of the cooling energy regression? CI on β_1 defines the uncertainty in load predictions.</i>	Resample dataset with replacement (10,000 bootstrap samples). Compute CI for regression slope and mean cooling energy. Plot bootstrap distribution. Interpret: what does the CI tell a designer? (<i>Lab independent — no Nov 27 class. Due before Mon Nov 30.</i>)	Lab Assignment 14 + Pre-Lab Quiz 15

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
15 Nov 30, Dec 2, 4	Final Exams + Capstone Work	No new lecture content this week.	Mon Nov 30 (50 min): Final Written Methods Exam — open notes, closed computer and AI; covers Weeks 9–14 concepts. Wed Dec 2 (50 min): Python Practicum — new dataset in Colab, open notes/documentation, no internet or AI; covers Weeks 9–14 Python skills. Fri Dec 4 (50 min): Capstone in-class work session — teams work on analysis; instructor available for questions.	Capstone Report due Sat Dec 6 by midnight (D2L)

Week	Theme	Lecture Topics	Lab / Workshop	Deliverable
16 Dec 7, 9	Capstone Presentations	No lecture — last two class days of the semester.	Mon Dec 7: ~9 teams present (4 min + 1 min Q&A each), brief transitions between teams. Wed Dec 9 (Last day of classes): ~6 teams present (4 min + 1 min Q&A each), followed by a full cross-team debrief once every team has presented. Every team chose its own dataset, so presentation order is by roster, not by shared topic.	Capstone Presentation

Lab Structure

Weeks 1–8 (Excel)

Each 50-minute lab session follows this structure:

1. **Briefing (5 min)** — introduce the dataset and engineering question for the session
2. **Setup (10 min)** — verify data loaded correctly; check column headers and units
3. **Core analysis (35 min)** — guided questions walk through the main computation; □ -marked questions require an intermediate result before the final answer
4. **Interpretation (15 min)** — short-answer questions connecting numerical results to engineering decisions
5. **Deliverable (15 min)** — 3-paragraph engineering memo exported as PDF; submitted to D2L

Weeks 9–15 (Python / Colab)

Each 50-minute lab session follows this structure:

1. **Setup (10 min)** — open Colab notebook from D2L link; run setup cell to import libraries and load dataset
2. **Guided code cells (25 min)** — scaffold cells with blanks to complete; students fill in function arguments and variable names

3. **Core model (15 min)** — run and inspect output (coefficients, p-values, percentiles)
 4. **Visualization (15 min)** — produce required plot with title and axis labels
 5. **Deliverable (15 min)** — add Markdown conclusion cell (3–5 sentences); export notebook as PDF via File □ Print □ Save as PDF; submit to D2L
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Policies

Pre-Lab Quizzes

Ten multiple-choice questions due on D2L **before Session 3 (Lab)** each week. Quizzes cover the prior week's lecture material and the Tutorial content from Session 2. Purpose: ensure students arrive at the lab having reviewed the relevant concepts. No late submissions accepted; lowest quiz score is dropped.

In holiday-shortened weeks (Weeks 3, 12, and 14), no in-class lab session is held; the quiz is due before Friday's Tutorial session.

Lab Assignments

- **Auto-graded component:** numeric answers checked against key with tolerance (typically $\pm 1\%$). Submitted through D2L.
- **Manual component:** charts and 3-paragraph memos graded by instructor/TA on a rubric distributed with each assignment.
- No late lab assignments accepted. All lab submissions must be made by the posted deadline; no exceptions.

Lab Template Policy

Each lab uses a provided answer sheet (Excel, Weeks 1–8) or scaffold notebook (Python, Weeks 9–15). The auto-grader reads answers from specific cells or output variables; work placed outside those locations receives **zero credit**.

You must not: - Rename or re-tab the answer sheet file - Move, insert, or delete rows in the answer sheet - Place answers in the wrong column - Rename `ANSWER_` variables in Python notebooks

One-time fix window: If you realize your submission deviated from the template, email a corrected file to the instructor within **24 hours** of the deadline. After that, no manual re-grading is provided and zero auto-graded credit stands.

Exams

- **Midterm (Week 8):** Session 1 — Concept Exam (closed computer, calculator permitted, 50 min). Session 2 — Lab Practicum (open Excel, open notes, no internet or AI, 50 min).
- **Final Exam (Week 15, 25% of course grade):**
 - **Session 1 — Written Methods (Mon Nov 30):** Open notes, open textbook, closed computer and AI. 50 minutes. Covers Weeks 9–14 concepts.

- **Session 2 — Python Practicum (Wed Dec 2):** Open notes and Python documentation. Google Colab only — no internet browsing or AI tools. New dataset not seen before the exam. 50 minutes. Covers Weeks 9–14 Python skills.
- No make-up exams except under documented extraordinary circumstances coordinated through the Dean of Students Office.

Capstone Project

- Teams of three assigned by instructor at the end of Week 13 (Fri Nov 20 lab session) — ~15 teams total.
 - Each team chooses its own real-world civil/architectural engineering dataset — any topic, any public source, no instructor approval required (see `Wk15.16_Dataset_Selection_Guide.md` for guidance and suggested sources). Five ready-made building-energy datasets remain available as an option for teams who prefer not to search. All three members of a team analyze their team's chosen dataset independently in their individual lab notebooks.
 - Plagiarism between teams (sharing results, copying analysis) is treated as academic dishonesty.
 - **Report due:** Sat Dec 6 by midnight (D2L submission) — one report per team, 40 raw points, distinct from the individual lab notebook (100 pts) and the team presentation (60 pts). All three deliverables are scaled into the Capstone Project + Presentation category.
 - **Presentations:** Mon Dec 7 and Wed Dec 9 (last two class days); 4 minutes per team plus 1 minute Q&A, all members must present a portion.
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Attendance and Participation

Attendance at both weekly sessions (lecture and lab) is expected. Participation in labs is evaluated through submitted lab assignments; there is no separate participation grade for lecture attendance.

- If you are ill or may have been in contact with someone infectious, stay home and notify the instructor by email before the session.
- Non-attendance does not guarantee automatic extensions or exam rescheduling.
- If you must miss the equivalent of more than one week of class, contact the Dean of Students Office: DOS-deanofstudents@arizona.edu or 520-621-2057.

Pre-Approved Absences

Students participating in UA-sanctioned activities (intercollegiate athletics, ROTC field exercises, Model UN, etc.) that conflict with class must provide documentation from the sponsoring organization to the instructor within the first two weeks of the semester. Arrangements for missed work will be made in advance.

Classroom Behavior

Students are expected to treat all class participants — instructor, peers, and guests — with respect. Behavior that disrupts the learning environment, including but not limited to persistent off-task device use, intimidating or disrespectful communication, and audio or video recording

without the instructor's consent, is not permitted. Students who engage in disruptive behavior may be asked to leave the session. Threatening behavior will be reported to the UA Behavioral Intervention Team in accordance with the [UA Threatening Behavior by Students Policy \(ABOR 5-308\)](#).

Use of AI Tools

You are encouraged to use AI tools as a learning aid — to explain concepts, debug code, and check reasoning. Example uses will be demonstrated in class.

What AI cannot replace in this course: forming your own engineering judgment from data. The lab assignments and exams require you to interpret results and make design recommendations. An AI that produces the number does not earn credit; you must show the reasoning.

Attribution requirement: Any work that used AI assistance must be acknowledged with a brief note (e.g., “Used Claude to debug the COUNTIFS formula in A3”). Submitting AI-generated analysis without attribution is academic dishonesty under the UA Code of Academic Integrity.

Note: AI apps and web searches are **not permitted** during exams. Exams allow: your notes, the course materials, Excel or Python, and calculator-approved apps.

University Policies

A comprehensive list of university syllabus policies is available at academicaffairs.arizona.edu/syllabus-policies. The following policies are specifically required to be included in this syllabus.

Academic Integrity

The University of Arizona Code of Academic Integrity applies to all work in this course. One permitted sanction for dishonest academic work is a failing grade (E) for the course. Sharing lab answers, copying code, and submitting AI output without attribution are all violations. See the full policy at deanofstudents.arizona.edu/academic-integrity.

Nondiscrimination and Anti-Harassment

The University of Arizona is committed to creating and maintaining an environment free of discrimination and harassment based on race, color, national origin, sex, age, disability, religion, sexual orientation, gender identity, genetic information, veteran status, or any other characteristic protected by law. Students who believe they have experienced or witnessed discrimination or harassment are encouraged to contact the Office of Institutional Equity at equity.arizona.edu or 520-621-9449.

Disability Accommodations

Students with disabilities who require reasonable accommodations to participate in this course must register with the Disability Resource Center (DRC) at drc.arizona.edu or 520-621-3268. Stu-

dents should provide a current accommodation letter from the DRC to the instructor within the first two weeks of the semester. Accommodations are not retroactive.

Religious Observance

Students who wish to observe religious holidays that conflict with a class session, quiz, or exam must notify the instructor in writing within the first two weeks of the semester. The instructor will make reasonable accommodations wherever possible. The full university policy is available at policy.arizona.edu/religious-accommodation-policy.

Threatening Behavior

The University of Arizona takes threatening behavior seriously. Students who engage in threatening conduct will be reported to the UA Behavioral Intervention Team (UABIT) in accordance with the [Threatening Behavior by Students Policy \(ABOR 5-308\)](#). Students who feel threatened or observe threatening behavior should contact the UA Police Department (520-621-8273) or the Dean of Students Office.

Pre-Approved Absences

See the Attendance and Participation section above for the pre-approved absences policy applicable to this course.

University Resources

Academic advising: advising.arizona.edu · 520-626-8667 · advising@arizona.edu

Campus Health (medical): 520-621-9202 · After Hours: 520-570-7898

Counseling & Psych Services (CAPS) 24/7 hotline: 520-621-3334

Dean of Students: DOS-deanofstudents@arizona.edu · 520-621-2057

Disability Resource Center (DRC): Students requiring accommodations must provide a current DRC letter. Contact the instructor within the first two weeks of class.

Concept Depth Map — Weeks 1–8

This table shows when each concept is introduced (I), fully developed (D), or reviewed at mastery level (M). Use it to calibrate how much time to spend on each topic.

Concept	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8
Mean, Median, Mode	D	M	M	—	—	—	—	Review

Concept	Wk 1	Wk 2	Wk 3	Wk 4	Wk 5	Wk 6	Wk 7	Wk 8
Range, SD, Variance	D	M	—	—	—	—	—	Review
Quartiles / IQR	D	M	—	—	—	—	—	Review
Skewness	I	D	M	—	—	—	—	Review
Histogram	—	D	M	—	—	—	—	Review
Box Plot	I	D	M	—	—	—	—	Review
Percentiles (design thresh-old)	—	D	M	M	—	—	—	Review
Conditional stats (AVERAGEIF)	D	M	M	—	—	—	—	Review
Classical probability / conditional	—	I	D	—	—	—	—	Review
Bayes' Theorem	—	—	D	—	—	—	—	Review
Poisson distribution	—	—	—	D	—	—	—	Review
Normal distribution	—	—	—	—	D	D	—	Review
Empirical vs. parametric	—	—	—	—	—	D	—	Review
Correlation / Pearson r	—	—	—	—	I	—	D	Review

Datasets Used in This Course

Weeks	Dataset	Source	Key Variables
1–2	Tucson, AZ hourly weather 2023	NOAA / TMY3	Temperature (°F), Wind (mph), Relative Humidity (%)
3	Tucson office building energy 2023	DOE EnergyPlus simulation	Cooling, Heating, Lighting, Plug (kWh/hr)
4	Tucson daily precipitation 2013–2022	NOAA	Daily precipitation (in), 10 years
5–8	DOE Medium Office zone temperature 2023	DOE EnergyPlus simulation	Zone temp, Outdoor temp, HVAC energy (8,760 hourly rows)
9–14	DOE Medium Office zone temperature 2023	Same as above	Same; used as primary Python dataset
15–16	Team-chosen — each capstone team selects its own real-world civil/architectural engineering dataset from any public source	Varies by team; see Wk15.16_Dataset_Selection for suggested sources	Varies by team. Five ready-made synthetic building-energy datasets (Hospital, PrimarySchool, LargeRetail, Warehouse, SmallHotel; EnergyPlus-calibrated) remain available as a no-search-required option.

All datasets are provided through D2L. Do not share datasets outside the course.

Subject to Change

The information in this syllabus is subject to change with reasonable advance notice. Changes will be announced in class and updated on D2L. Students are responsible for reading and understanding this document and for monitoring D2L for updates.

Exceptions: The grading scale and the absence/late work policies stated in this syllabus will not be changed after the first day of class.

This syllabus was distributed on the first day of class in accordance with UA Undergraduate Course Syllabus Policy. Last updated: Fall 2026.